

Qijun Tong

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Education

University of Electro-Communications

Tokyo, Japan

Ph.D. in Informatics

Oct 2025 - Sep 2028 (expected)

Supervisor: Prof. Satoshi Hara

Research interests:

- Machine learning theory
- Optimal Transport and Gradient flow
- Mathematical optimization and its applications

Keio University

Tokyo, Japan

Master of Engineering

Apr 2018 - May 2020

Supervisor: Prof. Kei Kobayashi

Relevant coursework: Bayesian statistics, Statistical learning theory, Information geometry

Teaching assistant experience: Mathematical statistics

Keio University

Tokyo, Japan

Bachelor of Engineering

Apr 2014 - May 2018

Supervisor: Prof. Kei Kobayashi

Relevant coursework: Mathematical statistics, Mathematical optimization, Measure theory, Probability theory, Differential geometry, Image Processing

Publications

Qijun Tong, Masahiro Ikeda, and Ryota Kawasumi. High-Probability Guarantees for SGD under β -Heavy-Tailed Gradient Noise. Submitted to ICLR 2027, 2026.

Qijun Tong and Kei Kobayashi. Entropy-regularized optimal transport on multivariate normal and q-normal distributions. Entropy, 23(3):302, 2021.

Preprints

Qijun Tong, Masahiro Ikeda, and Ryota Kawasumi. High-probability guarantees for SGD under β -heavy-tailed gradient noise. arXiv:2609.32195, 2026.

Talks

“High-Probability Guarantees for SGD under β -Heavy-Tailed Gradient Noise” (poster), YAML 2026, Yamagata, Japan, Sep 2026

“On the Knothe-Rosenblatt Distance between Gaussian Processes”, 2nd Workshop on Interdisciplinary Statistical Analysis through Computational Techniques, The Institute of Statistical Mathematics, Tokyo, Japan, Feb 2026

“Bernstein Processes and Their Applications”, Statistics Summer Seminar 2025, Kagawa, Japan, Aug 2025

“Matching Vector Images for Automated Inbetweening in 2D Animation Production”, Japanese Joint Statistical Meeting, Chuo University, Tokyo, Japan, Sep 2018

Fellowships & Grants

JSPS Research Fellowship for Young Scientists (DC1)

Apr 2026 - Mar 2029

JST SPRING Fellowship

Oct 2025 - Mar 2026

Professional experience

Data scientist/Optimization Engineer

Tokyo, Japan

Accenture Japan Ltd

Jun 2023 - Nov 2025

- Supervised the development of advanced scheduling algorithms for construction and retail clients, focusing on developing efficient mathematical optimization systems.
- Developed proper algorithm, utilized JIT and Cython for performance improvement, and engaged in ETL data modeling, deployment with AWS EKS, and CI/CD environments.
- Collaborated with various teams to define testing processes and develop maintenance and operational procedures, ensuring robust and efficient digital transformation solutions.

Data scientist/ML Engineer

Tokyo, Japan

ALBERT Inc.

Apr 2020 - May 2023

- Led the development and implementation of a social media analysis tool for a securities firm, focusing on designing the analysis architecture and managing data crawling for reputation analysis.
- Programmed and introduced a customized Business Intelligence (BI) system.
- Conducted R&D on natural language models to summarize customer complaints for a telecom company, enhancing models with the latest research and fine-tuning them using AWS resources for Proof-of-Concept development.

RIKEN Centre for Advanced Intelligence

Research Assistant (Part-time)

Apr 2019 - Mar 2020

- Involved in the research related to the RIKEN Centre's Mathematical Science team project, assisting with research projects, preparing for conferences, and participating in seminars.

Languages

Japanese (Native), English (Intermediate, TOEFL 92, IELTS 7.0), Chinese (Conversational)

Skills

Mathematical foundations: a solid foundation in optimization theory, functional analysis, and probability theory.

Technical skills: Experienced in data processing, machine learning, and mathematical optimization.

Programming Languages

Python, C++, Rust

Libraries & Tools

PySpark, PyTorch, Cython, Scikit-learn, OpenMP, SIMD (AVX), CUDA, CuPy, MySQL, PostgreSQL

Software & Frameworks

Microsoft Office, Photoshop, Git, Redmine, Argo Workflow

Cloud Platforms

Amazon AWS, Microsoft Azure

Miscellaneous

Competitive programming: Atcoder, Codeforces (approximate top 15%)

Digital illustration: Gallery